

Computational Breast Anatomy Simulation Using Multi-Scale Perlin Noise

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Abstract—Virtual clinical trials (VCTs) of medical imaging require realistic models of human anatomy. For VCTs in breast imaging, a multi-scale Perlin noise method is proposed to simulate anatomical structures of breast tissue in the context of an ongoing breast phantom development effort. Four Perlin noise distributions were used to replace voxels representing the tissue compartments and Cooper's ligaments in the breast phantoms. Digital mammography and tomosynthesis projections were simulated using a clinical DBT system configuration. Power-spectrum analyses and higher-order statistics properties using Laplacian fractional entropy (LFE) of the parenchymal texture are presented. These objective measures were calculated in phantom and patient images using a sample of 140 clinical mammograms and 500 phantom images. Power-law exponents were calculated using the slope of the curve fitted in the low frequency $[0.1, 1.0] \text{ mm}^{-1}$ region of the power spectrum. The results show that the images simulated with our prior and proposed Perlin method have similar power-law spectra when compared with clinical mammograms. The power-law exponents calculated are -3.10 , -3.55 , and -3.46 , for the log-power spectra of patient, prior phantom and proposed phantom images, respectively. The results also indicate an improved agreement between the mean LFE estimates of Perlin-noise based phantoms and patients than our prior phantoms and patients. Thus, the proposed method improved the simulation of anatomic noise substantially compared to our prior method, showing close agreement with breast parenchyma measures.

Index Terms—Perlin noise, virtual clinical trial, anthropomorphic breast phantom, digital breast tomosynthesis.

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I. INTRODUCTION

VIRTUAL clinical trials (VCTs) for the evaluation, optimization, and validation of novel imaging systems require computational simulations of human anatomy (phantoms) [1]–[8]. These phantoms have been used widely in breast imaging VCTs to support the pre-clinical assessment of mammography systems [9]–[11]. A VCT requires representative phantoms of a specific population with and without disease conditions and pathologies [1], especially when the VCT is targeted toward a particular clinical task [8]. For instance, breast imaging VCTs should simulate the anatomic noise seen in images of the mammary parenchyma, in particular for applications that require characterization of breast lesions or estimation of risk of masking cancers. Parenchymal texture is highly correlated to masking risk [12]–[14], affecting the ability of observers to detect and characterize breast abnormalities [15].

Digital mammography (DM), including digital breast tomosynthesis (DBT), has evolved rapidly over the last decade, resulting in improved visualization of the mammary parenchyma and increased sensitivity of cancer detection [16]–[19]. VCTs offer a powerful *in-silico* alternative to accelerate the development of novel imaging technologies. VCTs should be able to replicate clinical trial studies by demonstrating the magnitude of improvement in imaging modalities. VCTs should also be sensitive to changes in diagnostic performance using new or modified imaging modalities. In x-ray breast imaging, many researchers have been using VCTs to predict improvements in DBT over DM [2], [6].

There is an ongoing effort by VCT researchers to improve the realism of human breast anatomy phantoms [15], [20]–[23]. We have been using anthropomorphic breast models developed at the University of Pennsylvania in several breast-imaging applications [2], [24]–[30]. These anthropomorphic phantoms simulate anatomic structures of the breast including: skin, adipose and fibroglandular tissue, and Cooper's ligaments [10], [31]. However, depending upon the selection of simulation parameters, the resulting images may contain visible geometric structures [24], [32] that are considered unrealistic by human observers. It may also be possible to construct phantoms that are deemed to be realistic by humans but differ in higher-order statistics that would be perceived by machine observers.

Perlin noise has been used to simulate natural looking images, such as computer-rendered scenes in

computer-generated imagery (CGI) [33]–[37] (e.g., clouds, fog, and water), and texture or patterns in medical imaging [38]–[42]. Perlin noise uses random gradient values that are smoothly connected by an interpolation function to simulate complex textures [43]. The simulation of Perlin textures depends on the combination of scaled noise functions that are implemented on a spatial frequency band with octave bandwidth [44], [45].

A fractal implementation of Perlin noise has been developed to simulate different amounts of breast density for anthropomorphic phantoms [38], [39]. The proposed method builds upon this work by dividing the breast tissue into regions with different noise characteristics to simulate glandular and adipose tissues more realistically. We routinely use human readers to tune and evaluate subjectively the realism of the simulated parenchyma. However, human reader studies must necessarily be limited in extent due to concerns of reader fatigue and time constraints. Thus, objective measures should be considered when one must optimize a large number of simulation parameters.

Power spectra measurements have been extensively explored to investigate, compare, and characterize breast images [30], [46]–[48]. However, these measurements do not fully characterize anatomical noise variability, since higher-order statistical structures are not necessarily captured in a second-order statistic [49], [50]. In a previous study, we proposed the use of the power spectra and Laplacian fractional entropy (LFE) measures to evaluate higher-order statistical properties of phantom images objectively [30]. The LFE measure is calculated from the relative entropy of the response histogram of an image compared to that of a Gaussian histogram matched for mean and variance [50]. These measures have shown to be an effective method to evaluate phantom realism and to select simulation parameters [30].

In this paper, we present a new method to simulate fine structures of breast tissue. The shape and gross anatomy are based on our existing breast phantom software [51], however the relatively uniform breast tissue compartments and Cooper's ligaments are replaced with mixtures of glandular and adipose voxels. The phantom is divided into four layers (masks) and a unique Perlin noise distribution is created for each layer. A multi-scale approach was used to define the Perlin noise between layers. The resulting Perlin noise was then used to introduce fine-scaled anatomical noise in the breast phantom. DM and DBT projections of the breast phantoms are synthesized using computer simulations of a clinical system. Finally, power-spectrum profiles and LFE measures are used to assess the breast phantom images through comparison with clinical mammograms.

II. MATERIALS AND METHODS

A. Perlin Noise

Perlin's method involves the combination of multiple, scaled versions of a noise function [44], [45], [52]. The noise function is calculated from a discrete sequence of random gradient values g_u ($u \in \mathbb{Z}$); the gradients are uniformly distributed in the $[-1, 1]$ interval [43]. The values of the noise function (N)

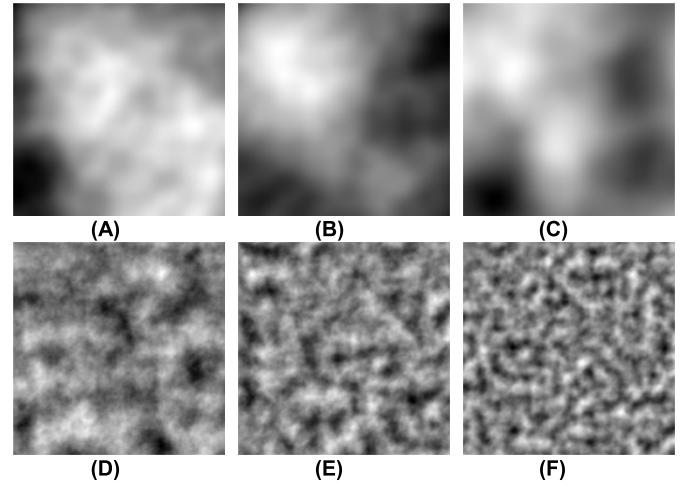


Fig. 1. Examples of Perlin noise frequencies (2D) simulated using respective coefficients f_{min} , K , and Φ : (A) 0.001, 4, 0.5; (B) 0.002, 3, 0.4; (C) 0.003, 2, 0.3; (D) 0.01, 4, 0.5; (E) 0.02, 3, 0.4; and (F) 0.03, 2, 0.3.

are obtained by piecewise interpolation using a local interpolating function F at arbitrary positions $x \in \mathbb{R}$ following the formalism:

$$N(x) = F(x, g_u, g_{u+1}), \quad \text{with } u = \lfloor x \rfloor; \quad (1)$$

The Perlin noise function (PN) is defined as the linear combination of multiple scaled version of the same noise function with K different frequencies and amplitudes: [43]

$$PN(x) = \sum_{i=0}^{K-1} a_i \cdot N(f_i \cdot x) \quad (2)$$

The frequencies are not arbitrary; i is an “octave” step. The frequencies are successively doubled in each octave step in the form $f_i = 2^i f_{min}$, for $i = 0, 1, \dots, K-1$, where K represents the total number of frequencies [53], [54].

The amplitude values vary with the power of i in the form $a_i = \Phi^i$. The coefficient Φ is a constant that represents the “persistence” and it is typically set in a $[0.25, 0.5]$ interval [43]. The properties of the Perlin noise depend essentially on the selection of the coefficients Φ , f_{min} , and the number of noise frequencies (K). A detailed formalism of the N -dimensional Perlin noise is found in Burger and Burge [43].

The noise functions are implemented in Java and distributed using a common library containing a collection of ImageJ plugins [55]. The common library was integrated into our open source Virtual Clinical Trial (VCT) framework [51], [56]. The VCT framework uses multithreaded processing on 3D phantoms through slice-wise 2D Perlin noise simulations. The slice-wise approach accelerates phantom creation and reduces computational burden from the 3D Perlin noise simulation. The Perlin noise library includes a hashing function [43], which ensures that there is textural continuity between slices. Thus, it is not necessary to use a 3D noise generator. Fig. 1 shows example regions of interest (ROIs), simulated using the implementation of 2D Perlin noise [43]. These examples are representative of the simulation parameters used in this work.

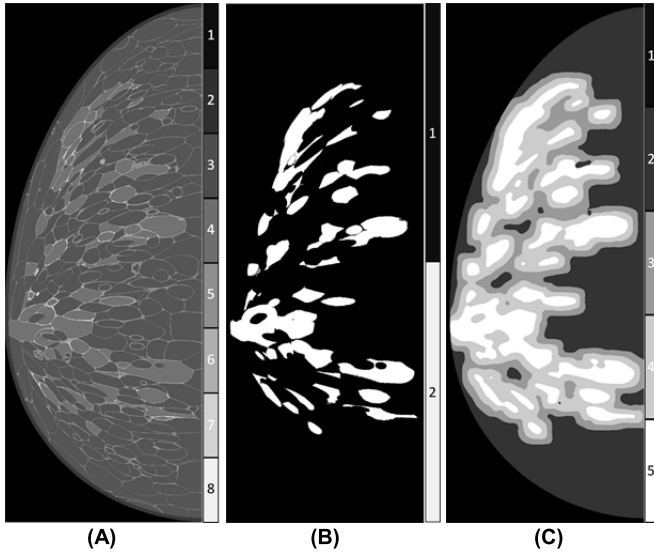


Fig. 2. (A) Central slice (sagittal plane) of one anthropomorphic breast phantom, which contains a complex combination of background (1) and breast tissue structures (2-8). (B) Glandular binary mask (2) used for morphological operations and (C) resulting adipose (2-3) and glandular (4-5) masks used for Perlin noise simulation.

B. Anthropomorphic Breast Phantoms

The VCT framework [51], [56] was used to generate voxelized breast phantoms [10], [31]. The phantoms are created using an octree-based recursive partitioning algorithm [10] that simulates anatomical structures of the breast. The algorithm sets compartment seeds randomly within a semi-ellipsoidal phantom model. Using the seed locations, ellipsoidal compartments split recursively, and fibrous Cooper's ligaments are set to connect the gap between compartments. Finally, a beta function is used to determine the composition of each breast compartment.

In total, 500 voxelized phantoms [(0.1 mm)³ voxel size, 700 mL volume, and uncompressed breast thickness of 126.6 mm] were simulated. The phantom compartments consist of two fundamental voxel materials: adipose and glandular tissue. Each compartment is connected by Cooper's ligaments ([0.1, 0.18] mm thick) and filled with various tissue compositions ({0.1, 10, 20, 30, 50}% dense) [31]. Breast compression was simulated using GPU accelerated mesh software [1], [57] to a medio-lateral (ML) compression of 50% (63.3 mm thick).

A sample breast phantom is shown in Fig. 2A. Major modifications in the breast phantom generation software were implemented in this study to simulate more realistic breast compartments and Cooper's ligaments.

1) Breast Tissue Masks: The breast phantoms are composed of voxel material labels that represent the ground-truth for various tissue types and air (Fig. 2A). Glandular tissue labels were used to create glandular binary masks (Fig. 2B). Two morphological operations of dilation were applied on the binary masks using the methods implemented in the Imglib2 library developed in Java [58]. The dilation operations expand and create glandular tissue mask layers, which connect breast compartments and reduce the geometrical appearance of

the dense compartments. We used dilations of 20 and 60 voxels with disk structuring elements. The output of each dilation mask and the original binary mask were combined, resulting in a three-layer dilation mask. Skin and background voxels were used to set the boundaries of the dilation mask. Dilated voxels located at the skin or in the background were cropped out from the dilation mask. Similarly, voxels that represent adipose compartments and adipose ligaments were identified to create a fourth binary mask for the adipose regions. This binary mask was combined with the dilated masks. The final resulting mask contains four layers; a different multi-scale Perlin noise distribution was calculated for each layer (Fig. 2C).

2) Perlin Noise Simulation: For each of the four mask layers, unique values of f_{min} , K , and Φ were selected based upon the work described by Burger and Burge [43] and values selected manually in a preliminary study we conducted with human observers. Thus, four independent distributions of Perlin noise were created to modify the voxels of the breast phantom.

For the most glandular layer, Perlin noise was simulated using f_{min} , K , and Φ in ranges [0.001, 0.007], [2, 4], and [0.2, 0.5], using steps of 0.001, 1, and 0.1, respectively. For the most adipose layer, Perlin noise was simulated using f_{min} , K , and Φ in ranges [0.01, 0.05], [2, 4], and [0.2, 0.5], using steps of 0.005, 1, and 0.1, respectively. For the two intervening layers, the parameter f_{min} was increased in equal steps from the glandular to the adipose layers, while Φ and K have the same values assigned to the most glandular layer.

The Perlin noise was normalized using a [1, n] interval, where $n \in \mathbb{Z}^+$ and represents the number of voxel types assigned for the breast phantom. In this study, we considered $n = \{4, 8, 16, 32, 63\}$ to normalize the noise distributions. Thus, the voxels that represent the binary compartments in the prior breast phantoms (i.e., either adipose or glandular) were replaced to create multi-scaled voxels.

We varied each voxel based on the normalized Perlin noise distributions using a complementary mixture of adipose and glandular tissue (partial volume) [2], [27]. For voxels located at the most adipose layer (Fig. 2C.2), the adipose composition decreases from 99.9% to 50%; for the most glandular layer (Fig. 2C.5), the glandular composition increases from 50% to 99.9%. Similarly, for the partial volume of voxels located at the two intervening layers (Fig. 2C.3 and 2C.4), the percentage of glandular composition increases from 50% to 99.9%. A total of n partial volume combinations are simulated. The increase and decrease of the percent of tissue composition in the partial volume of each distribution follow two functions of n – linear ($an + b$) and hyperbolic tangent [$\tanh(4n - 2)/2 + 0.5$]. Each value of n represents a particular voxel in the modified phantom (Perlin noise phantom) and it corresponds to a unique x-ray attenuation on the simulated images [59].

Finally, the voxels that represent Cooper's ligaments in the prior breast phantom were added back to the Perlin noise phantom. The x-ray attenuation of these voxels was also modified using partial volume of adipose and glandular tissue. We increased the portion of glandular composition by 20% to 50% and decreased the adipose composition accordingly in the Perlin noise phantoms.

TABLE I
SUMMARY OF IMAGING ACQUISITION GEOMETRIES SIMULATED

X-ray Modality	DBT	DM
Anode Material (a.u.)	W	W
Filter Material (a.u.)	Al	Rh/Ag
Filter Thickness (mm)	0.7	0.05
Angular Range (°)	[-7.5, +7.5]	-
Number of Projections (#)	15	1
Tube Motion (a.u.)	Continuous	Stationary
Detector		
Detection Material (a.u.)	a-Se	a-Se
Detector Element Size (mm)	0.14×0.14	0.07×0.07
Number of Elements (#)	2048×1664	4096×3328
Detector Size (mm)	286.72×232.96	286.72×232.96
Source-to-Image Dist. (mm)	700.00	700.00
Reconstruction		
Pixel Size (mm)	0.10×0.10	-

C. Image Acquisition Simulation

DM and DBT projections of the breast phantoms were simulated using the OpenVCT Manager module [51], [56], which executes a fast GPU implementation of the Siddon algorithm for ray-tracing [60]. The Siddon algorithm traces a ray model from the source, through the phantom volume, onto each detector element. The radiological path length is determined by the sum of the length traveled by this ray in each voxel, multiplied by the density-ray attenuation of each voxel. This process is repeated for each ray trajectory, and the intensity of the rays in a detector element determine the pixel value for the projection image.

The DM and DBT projections were synthesized assuming the acquisition geometries shown in Table I. The acquisition exposure settings were simulated from automatic exposure control data of a clinical system [61]. We simulated polychromatic x-ray spectra. For DM images, the x-ray spectrum is simulated using W anode and Ag or Rh filtering materials. For DBT images, the x-ray spectrum is simulated using W anode with Al filtering materials. The attenuation coefficient data come from the ICRU Report 44 [62]. Reconstructed DBT slices were produced using simple backprojection with commercially available software library [63], [64].

D. Objective Assessment

A sample of clinical mammograms was used to collect objective measures of power spectrum and LFE. The clinical mammograms were acquired on Selenia Dimensions DBT systems (Hologic Inc., Bedford, MA). In total, 35 patients having 140 “for processing” screening mammograms (CC and MLO breast views). Similarly, DBT exams (CC and MLO breast views) were collected using a subset of these 35 patients (n=26). Nine patients were excluded because of incomplete data. In total, 104 DBT reconstructions were used for analysis [50]. Both DM and DBT clinical images were used as a reference for this study. The clinical data were collected retrospectively at UC Davis Medical Center and the Perlman Center for Advanced Medicine under IRB-approved human-subject protocols and HIPPA-compliant

non-identifiable exams. All clinical data were published and validated in a previous study [50].

Binary masks of the breast outline were provided and used to calculate power spectra and LFE measurements in regions of interest (ROI). These masks were used to exclude background (air) and a 10 mm boundary (chest wall and skin) of the clinical images. This exclusion criterion was also used in phantom images. A summary of methods used to collect sample ROIs and calculate the power spectrum and LFE measures are shown in Fig. 3.

1) **Power Spectrum:** Anatomical structure can be captured using power spectra measures collected in breast images [48], [49]. Burgess *et al.* have shown that these structures are characterized by a power-law function in the form κ/f^β , where f is the radial spatial frequency of the noise power spectrum, κ is the magnitude, and β is the power-law exponent of the image spectrum [48], [49].

In this study, we calculated and compared β estimates using an ROI of size 512×512 pixels (35.84 mm^2), extracted from each DM phantom image and clinical mammogram. We used the center of mass calculated on the binary masks to select ROIs within the breast or phantom images (Fig. 3A). The center of mass is used as a reference point to automate the process of cropping ROIs, ensuring only valid values (nonzero) to estimate power-spectrum (PS). The PS is estimated using the fast Fourier transform (FFT) of each ROI sample:

$$PS(u, v) = \frac{1}{N} \sum_{i=0}^{N-1} |FFT[w(x, y) * (r_i(x, y) - \bar{r}(x, y))]|^2 \quad (3)$$

where $r_i(x, y)$ represents a ROI sample, $\bar{r}(x, y)$ is the average of samples, and $w(x, y)$ is a spatial window that is centered at the ROI. This spatial window remains constant out to a radius of 8.96 mm before decreasing to zero with a cosine profile at a radius of 17.92 mm. The spatial window is used to reduce spectral leakage in the power spectrum estimates. The spatial frequency indices (u, v) are both in the interval $[0, N-1]$. This method was validated and published in a previous study [50].

We calculated the logarithm (base 10) of PS over the radial spatial-frequency interval to plot the profile of the power spectrum. The radial frequencies were binned using a bin width of size 0.0274 cycles per mm $[(1/(\text{ROI size} \times \text{pixel size}))]$ [50]. The standard deviation across every sample was used to quantify the variability of the log-power spectrum [50]. We used a total of 140 and 500 ROIs of mammograms and phantoms, resulting in a standard error of 8.5% and 4.5%, respectively.

To compare the power spectra of phantom and patient images, β estimates of the power spectra were calculated using the slope of the linear regression of frequencies from 0.1 to 1 cycles per mm [30], [49].

2) **Laplacian Fractional Entropy:** LFE has been used to evaluate statistical-properties of phantoms [30] and clinical images acquired or simulated using various breast imaging modalities [50] (e.g., computed tomography, DM, and DBT). The LFE metric is based on Gabor-filter response histograms resulting from 2D convolutions of Gabor kernels with an

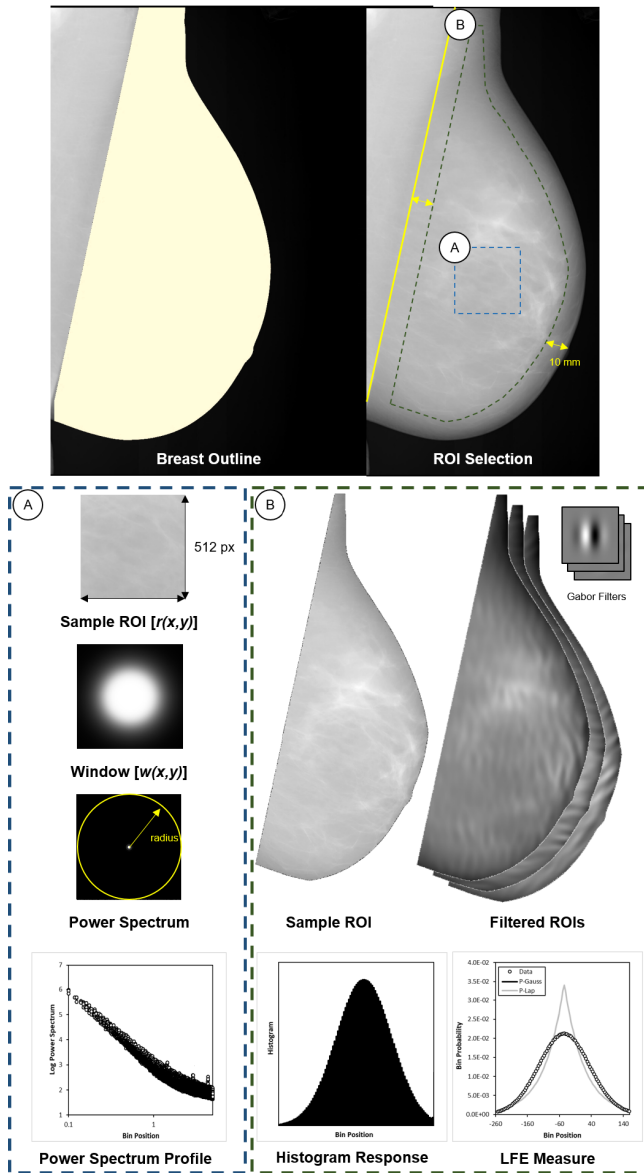


Fig. 3. Summary of method used to evaluate x-ray breast images objectively. (A) Assessment of the image power spectrum, and (B) Laplacian fractional entropy.

image in a region of interest (Fig. 3B). The measure is calculated from the relative entropy of the response histogram and that of a Gaussian histogram matched for mean and variance [30]. The result is normalized by the relative entropy of a matched Laplacian distribution, meaning that an LFE value of 100% is as different from a Gaussian distribution (in entropic terms) as a Laplacian distribution. This metric evaluates higher-order structural features in images, since it is invariant to the mean and variance of the histogram and only sensitive to higher-order statistical properties [50].

A set of Gabor filters was used and the kernel parameters were varied in accordance with previous publications [30], [50]. The center frequencies were varied from 0.125 to 4.0 cycles/mm and evaluated at six different orientations (from 0° to 150° in 30° increments). We used a bandwidth of 1.4 octaves, aspect ratio of 1, and shift phase of

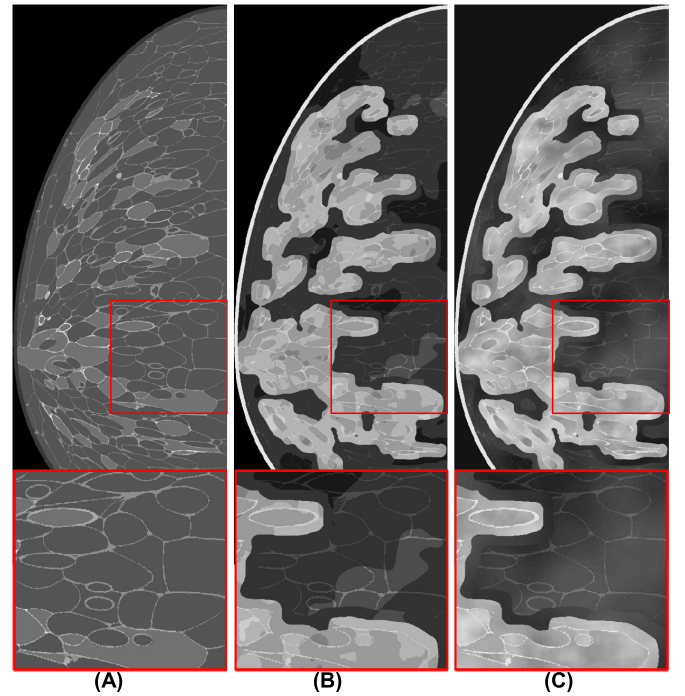


Fig. 4. Example of central slice (sagittal plane) of a breast phantom after processed using the normalized Perlin noise method. The distributions of Perlin noise were simulated using coefficients f_{min} , K , and Φ : 0.001, 3, 0.5 for the most adipose layer and 0.01, 3, 0.5 for the most glandular layer. (A) Prior phantom, Perlin phantom with (B) $n = 8$, and (C) $n = 63$.

$\pi/4$ radians. The histogram responses were obtained using 2D convolution through a selected ROI.

The response histograms were calculated in each filtered ROI and the LFE measures were calculated at different frequencies. The histograms were binned using 99% of responses, with an additional 1% bin for the remaining extremal values. The LFE was estimated using the methods specified in previous publications [30], [50].

Each DM projection was processed by log-converting the transmission values to density before LFE analysis [30], [50]. For DBT reconstructions, the LFE is calculated only in the central slice. No additional imaging processing was applied to the DM projections or DBT reconstructions.

The LFE results were averaged over the six Gabor filter orientations at each spatial frequency. For the clinical images, we averaged the LFE results over the 35 patients; separately, the result of the 500 phantoms were averaged. Based on previously reported results [30], [50], the standard error of the mean LFE (across breast views and patient exams) is relatively small (<4% of the mean on average). The LFE measures were compared using plot profiles, where mean and standard error are reported.

III. RESULTS

A. Perlin Noise Phantom

The Perlin noise and breast tissue masks modify the prior phantom to incorporate novel texture into the simulated mammary parenchyma (Fig. 4). Each normalized distribution of Perlin noise is used to assign a range of n voxel-materials.

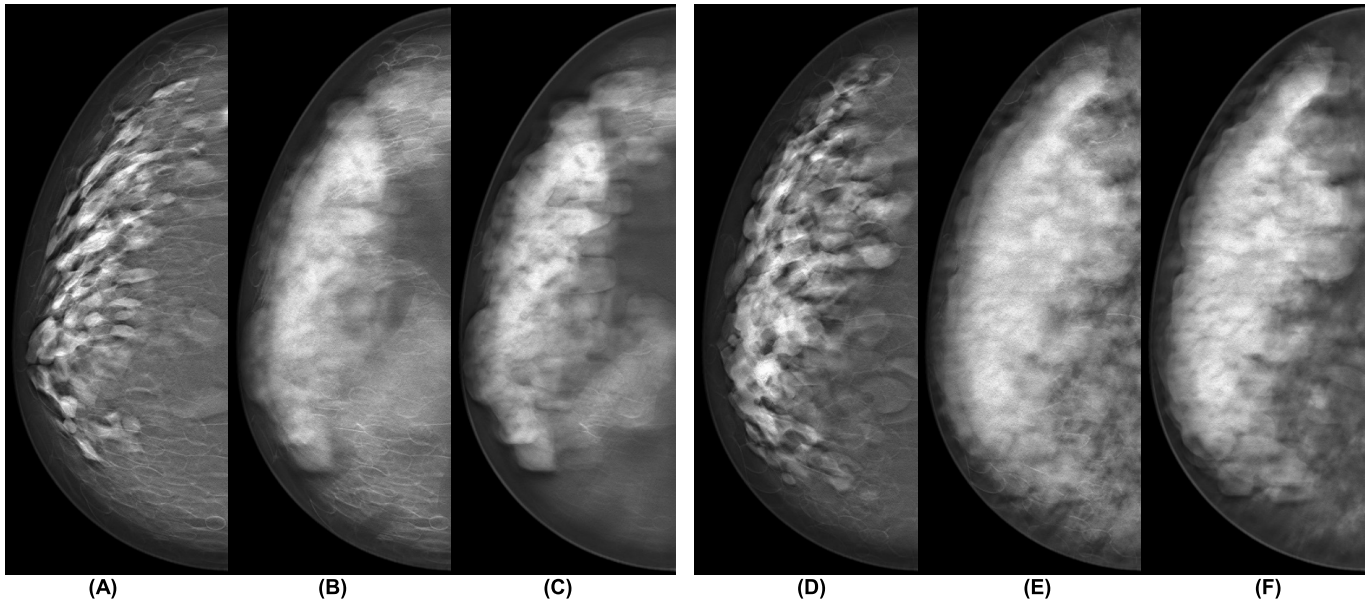


Fig. 5. Central slice (sagittal plane) of the DBT reconstruction of the two anthropomorphic breast phantoms used in this study. (A and D) prior reconstruction, and (B, C, E, and F) reconstruction using Perlin noise coefficients f_{min} , K , and Φ : $\{(0.001, 2, 0.4), (0.007, 4, 0.6)\}$ for the most adipose layer and $\{(0.05, 4, 0.6), (0.03, 5, 0.6)\}$ for the most glandular layer. The percent of breast tissue in the partial volume of each distribution follows the functions of $n = 63$: (B, and E) linear and (C, and F) hyperbolic tangent. For illustration, these particular images were processed using Briona (Real-Time Tomography, Villanova PA) imaging processing.

The n voxel-materials represent a different composition (i.e., partial volume combination) that vary in the fraction of glandular and adipose mixture. In this section, we report on $n = 63$ (Fig. 4C) to avoid unrealistic noise patterns in the simulated x-ray images.

Distinct x-ray attenuation properties are assigned for each voxel material. X-ray attenuation functions are used to control the attenuation and partial volume of adipose and glandular composition. Fig. 5 shows differences in the phantom images simulated using linear (Fig. 5B and E) and hyperbolic-tangent attenuation functions (Fig. 5C and F). These functions modify the overall breast density of the Perlin phantoms, and consequently, affect the contrast of the simulated x-ray images. In comparing images reconstructed using both functions, the hyperbolic-tangent function reduces the x-ray attenuation of the breast tissue substantially.

Note that our proposed method reduces the geometric features of the prior phantom compartments, resulting in Perlin texture that looks similar to anatomical noise structures found in the mammary parenchyma [30], [50]. Power spectrum and LFE measures were used to evaluate the phantom texture in simulated x-ray breast images.

B. Power Spectrum

The power spectrum profiles are shown in Fig. 6. The log-power spectrum was fitted using radial frequencies in the range 0.1 to 1 cycles per mm. The power-law exponents resulted in β values of -3.10 , -3.55 , and -3.46 , for the profiles calculated using clinical mammograms, DM images simulated with our prior phantoms, and DM images simulated with the proposed Perlin phantoms, respectively. The relative errors in β are within 10% of the sample estimate, which

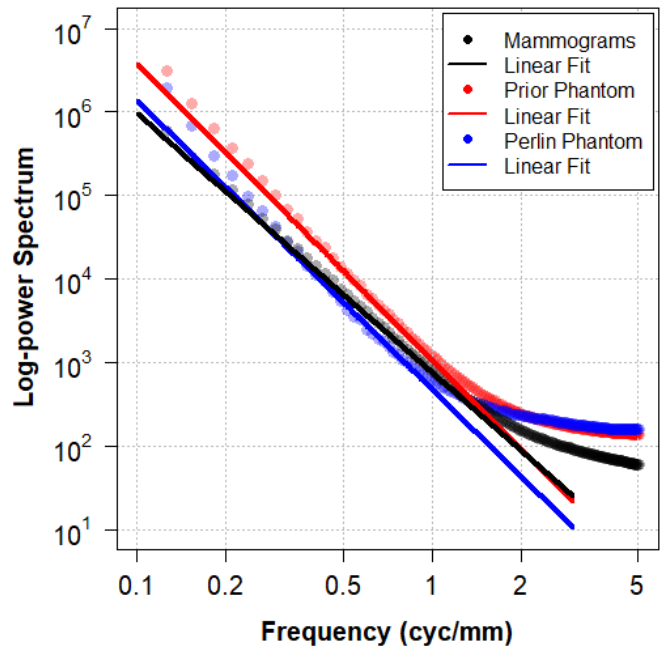


Fig. 6. Power spectrum profiles. These profiles are compared using simulated images and clinical mammograms. Linear regressions of the power spectrum are calculated across a frequency interval $[0.1, 1]$ cycles per mm to determine values of the power-law exponent (β).

is consistent with the previously reported results [30]. These results indicate that the power-law estimations of the proposed method are in agreement with the reference power spectrum calculated using clinical mammograms.

Note that these profiles suggest that both simulation methods are in a similar range of the power spectrum calculated

using clinical mammograms, even though the visual appearance of phantom and patient images are substantially different. These results illustrate the need for investigating additional objective metrics to evaluate statistical properties of Gaussian processes.

C. Laplacian Fractional Entropy

The LFE results for the various simulation conditions is graphed in Fig. 7. The general profile of the LFE measures is comparable to previous publications [30]. The simulated noise of the acquisition dominates the phantom signal at frequencies higher than 1 cycle/mm, driving the LFE results close to zero. The prior phantom shows substantial peaks in LFE measures across all frequencies that are not present in the clinical images. Note that the LFE computed using DM and DBT images simulated with Perlin phantoms have the closest agreement with the mammograms across the tested frequency range.

LFE shows substantial differences between the original phantom simulation procedure and the procedure that has been modified to include Perlin noise, and this comparison suggests that the proposed method produces more realistic textures in simulated images. Despite using parameters without separate optimization for Perlin noise (f_{min} , K , and Φ), the simulated DM and DBT images resulted in the closest agreement with the clinical data (Fig. 7A and Fig. 7B). These results support that the simulation procedure using Perlin noise more closely matches the LFE values from actual patient images.

The average LFE in the DBT images peaks slightly above the DM images in midrange frequencies from 0.25–0.5 cycles per mm and decays more rapidly to 0 above 1 cycle per mm [50]. In this study, the LFE distributions calculated using DBT images simulated with Perlin phantoms also peaks above the DM clinical images in the midrange frequencies and decays rapidly to 0 above 1 cycle per mm. However, more LFE analyses on clinical DBT images are recommended to support the use of the proposed methods. Ideally, LFE measures calculated in DBT images should consider the three-dimensionality of the reconstructions.

Examples of ROIs are shown in Fig. 8. ROIs cropped on clinical mammograms are used as reference (right column). Note that the DM images simulated with Perlin noise are comparable to the reference ROIs. The proposed method simulates noise texture for the phantom simulations (middle column), while the prior phantom shows overly distinct geometric compartments that are delimited by highly attenuated ligaments (right column). Ligaments and geometric compartments are “smoothed out” in images simulated using the proposed methods, supporting better LFE agreement between the anatomical texture of clinical images and the Perlin phantoms.

IV. DISCUSSION

We proposed a novel method to improve the simulation of anatomic noise and texture of anthropomorphic phantoms using multi-scale Perlin noise. The proposed method can be adapted, extended, and used in any phantom model if corresponding ground-truth of tissue labels is available. In this

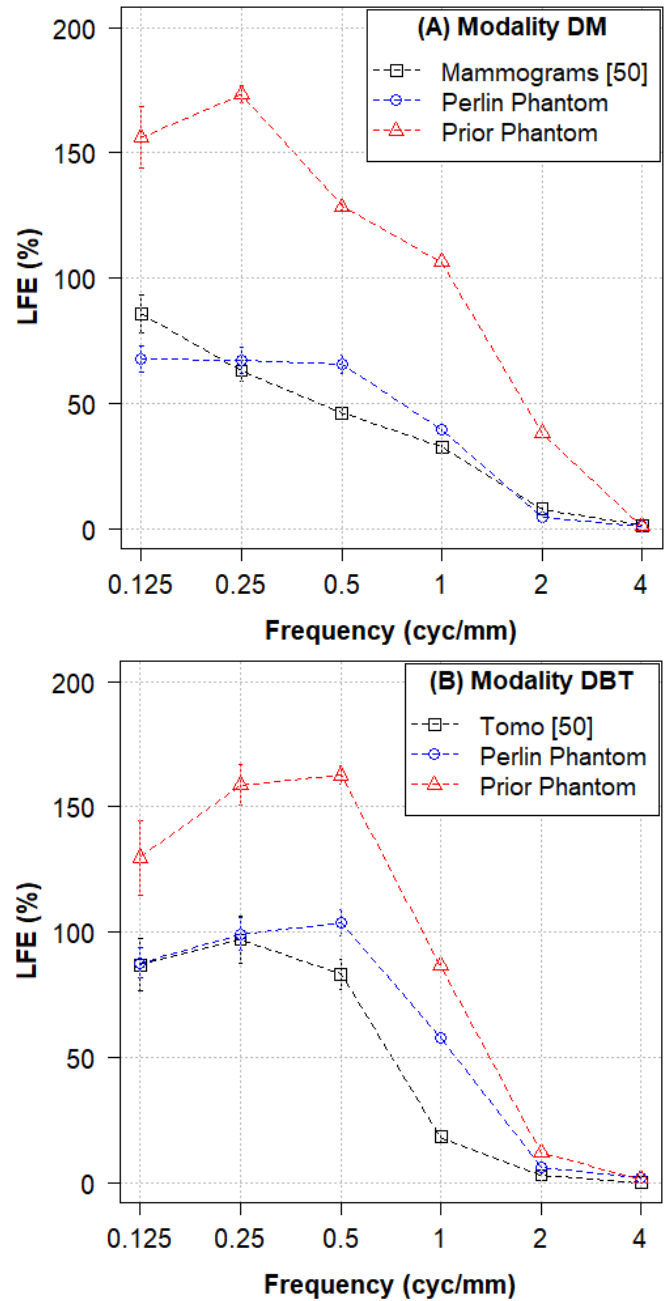


Fig. 7. Summary of LFE estimates (mean and standard error) collected using (A) DM and (B) DBT images. The LFE data for the clinical images (black box) is shown for reference [50]. The Perlin noise distribution (blue box) has the closest agreement with the reference clinical data across all frequencies.

study, we used breast phantoms developed in our laboratory that have been consistently revised, improved, and validated over the last 22 years [10], [31], [51]. Our current breast models use a recursive partitioning algorithm [10], which simulates tissue compartments bounded by Cooper’s ligaments. Depending upon the selection of simulation parameters, the Cooper’s ligaments can enhance the geometric appearance of adipose and glandular compartments, resulting in simulated images that may not be considered realistic enough for specific VCT tasks (e.g., VCT applications that require realistic simulation

of cancer masking or assessment of breast cancer risk). The proposed Perlin method reduces the notable appearance of breast compartments in the simulated images, by introducing anatomic noise and connecting tissues.

Our results show that the images simulated with our prior and proposed methods have similar power-law spectra when compared with clinical mammograms. The greatest differences in the power spectra are in the lowest (< 0.2 cycles per mm) and highest (> 2 cycles per mm) frequencies. These differences indicate that we can still optimize the simulation and acquisition parameters to match more closely the power spectrum of phantom and patient images. However, the power-law exponents (β) calculated were -3.10 , -3.55 , and -3.46 for patient, prior and Perlin phantoms, respectively. The β values show only a 10% difference in relative errors between patient and Perlin-noise based phantoms. These results support the data published previously [30] and confirm the need for higher-order statistical analyses to evaluate realism of simulated images.

The LFE metric was used to characterize non-Gaussian statistics in textures that was incorporated into our breast models. This metric has been suggested previously as a way to characterize non-Gaussian statistics in textures found in mammograms [30], [50]. That work demonstrated that LFE is relatively insensitive to the power spectrum of phantoms [51]. The LFE results in Fig. 7 show that the simulation procedure using Perlin noise more closely matches the values from actual patient images. This suggests that the Perlin-noise simulation more effectively captures the higher-order statistical properties of breast tissue. Since LFE is based on Gabor filters that are similar to the receptive fields in the early visual system [65], [66], these statistical properties should in principle be perceivable to human observers as well.

Our results show substantial LFE variability between Perlin and prior phantoms, phantoms and clinical mammograms, and between images simulated with different modalities (DM and DBT). The mean LFE of images simulated with Perlin noise provide closer agreement to the patient images compared with simulated images of our prior phantom. The LFE measures calculated using DM and DBT images with Perlin noise show a better agreement with the LFE calculated using patient images, especially at lower spatial frequencies of 0.125 and 0.25 cycles per mm that are dominated by anatomical variability. Based on these results, LFE has been demonstrated to be an authoritative metric for evaluating phantom realism objectively. This will allow us to evaluate modifications to our software, as well as the parameters are used to control and modify the texture appearance in the phantom images.

The combination of various simulation parameters affects and modifies the texture, overall density, and the realism of breast phantoms. In this study, we emphasized the following simulation conditions and parameters: (i) number and size of Perlin binary masks, (ii) range of Perlin frequencies, and (iii) x-ray attenuation functions. The simulation parameters used in (i) control and merge dense compartments, avoiding sharper and more geometric compartment edges in the prior phantoms. The number and range of Perlin frequencies

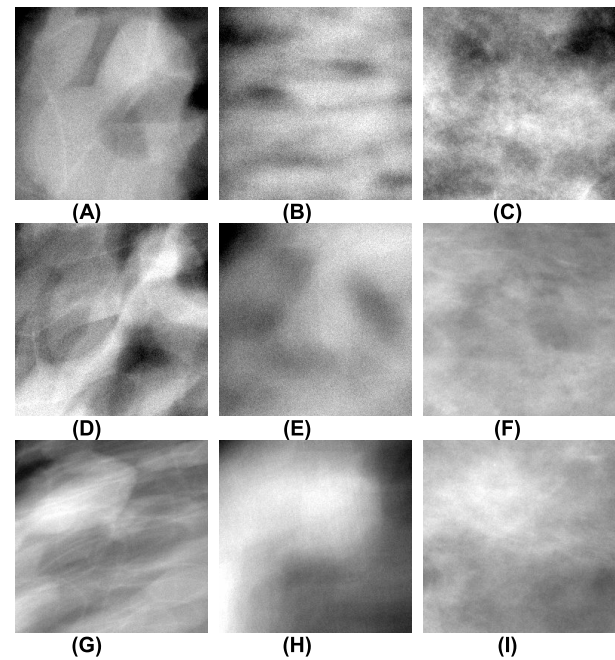


Fig. 8. Example of ROIs (256×256 pixels): (left) Prior phantom, (middle) Perlin phantom, and (right) clinical mammogram. The mean LFE ($\{0.125, 0.25, 0.5, \text{ and } 1.0 \text{ cyc/mm}\}$) results are: (A) {68.8, 76.8, 89.3, 79.11}, (B) {51.4, 40.0, 43.7, 31.9}, (C) {50.0, 39.9, 40.7, 30.6}, (D) {89.4, 88.8, 89.2, 82.7}, (E) {65.0, 36.5, 25.3, 19.3}, (F) {50.0, 36.5, 28.7, 19.8}, (G) {135.8, 146.1, 149.4, 118.8}, (H) {58.2, 57.2, 55.5, 28.2}, and (I) {65.4, 57.7, 44.1, 28.0}.

(ii) control the appearance of texture (Fig. 1) for each layer in the Perlin masks. Finally, the x-ray attenuation functions (iii) control the dense partial volume of each voxel-material. The hyperbolic attenuation function was selected to reduce the overall breast density and increase the density percent for the voxel-materials gradually. The simulation parameters should be carefully optimized and personalized for phantoms.

In this paper, the Perlin noise parameters (ii) were selected subjectively and applied to a diverse phantom population ($n = 500$). Although phantoms simulated with Perlin noise presented the closest LFE results to the sample of clinical images, the use of arbitrary parameters should be considered a limitation for this study. It will be beneficial to optimize the noise and phantom parameters using objective analyses, such as power spectrum and LFE. This optimization may further improve the phantom realism and match more closely the LFE and power spectrum of phantom images to the patient images. Thus, there is still some improvement to be made, both in terms of matching at higher and lower frequencies (below 0.2 and above 2 cycles per mm) and in terms of matching acquisition noise and breast density between patient and phantoms. Once the patient and phantoms are matched in terms of breast density, the LFE measures can be validated using visual interpretation of experts.

It is important to mention that the current study was not designed to match the most appropriate phantom image to a particular patient image. A more detailed phantom evaluation should consider a large clinical population of patients from which sample images are used. A larger sample of clinical DM

and DBT images is recommended to support the use of the proposed methods. Once the phantom and acquisition parameters are optimized and validated with larger clinical datasets, statistical significance tests should be performed to support the improvement in realism using Perlin noise phantoms.

Because DBT images reduce the superposition of breast tissue [16]–[19], we included samples of processed DBT reconstructions for the best visualization of differences in reconstructed slices simulated using both prior and Perlin phantoms. The examples provide some additional evidence that the proposed method improves the computer simulations, including anatomic noise in the simulated mammary parenchyma and reducing the effect of geometrical breast compartments in simulated images (Fig. 5A and D). The addition of more anatomical features will also help to reduce the geometric appearance of the phantom.

Our multi-scale Perlin method may eliminate the need of generating and combining phantoms (i.e., prior method [31]) to improve the phantom realism. In addition, our proposed method potentially can improve the algorithm performance for computer simulations. The software performance can be improved further by implementing the proposed methods using GPU-programming to accelerate the simulation of Perlin textures.

Finally, our VCT software inserts lesions using voxel replacement [31]. This method replaces phantom voxels with calcium (calcification clusters) or admixtures of glandular and adipose tissue (soft tissue masses). Voxel replacement for calcified lesions is very straightforward; calcium is highly attenuating. However, the simulation of masses is more difficult, especially in regions of uniform glandular tissue, because there may be no differentiating feature of the mass. Based on our observations, we believe that masses would be better differentiated from the parenchymal tissue based on differences in texture. We will investigate the simulation of masses with Perlin noise in a future publication.

V. CONCLUSION

This work describes a method to improve the realism of anthropomorphic phantoms. We propose the use of a VCT method that incorporates Perlin noise structures into anthropomorphic phantoms. Images simulated with Perlin-noise based phantoms have more similar power-law spectrum and power-law exponents than our previous phantoms when compared to mammograms; higher-order statistical properties of the Perlin-noise based phantoms, as measured by LFE, overlap the statistical properties measure in patient images.

The samples of simulated images presented in this study show that the proposed method improves the simulation of breast tissue, where Copper's ligaments and geometric compartment shapes are less evident in the simulated x-ray phantom images. These results suggest the use of Perlin-noise based phantoms for VCT applications that require the most realistic simulation of mammary parenchymal tissue.

Recruiting human readers to inspect images and score realism can be expensive, and time-consuming. The proposed validation method using power spectrum and LFE gives us a

way to quantify improvements in realism objectively. Based on these results, we will use power spectrum and LFE as a primary tool for evaluating our future phantom design changes. We encourage the broader use of objective metrics, such as power spectrum and LFE, to evaluate phantom realism and to optimize simulation parameters in anthropomorphic x-ray simulations.

REFERENCES

- [1] P. R. Bakic, B. Barufaldi, D. Pokrajac, M. A. Lago, and A. D. Maidment, "Developing populations of software breast phantoms for virtual clinical trials," in *Proc. 14th Int. Workshop Breast Imag. (IWBI)*, Jul. 2018, p. 73.
- [2] P. R. Bakic *et al.*, "Virtual clinical trial of lesion detection in digital mammography and digital breast tomosynthesis," *Proc. SPIE*, vol. 2018, Mar. 2018, Art. no. 1057306.
- [3] T. Vent, R. Acciavatti, and A. Maidment, "Development and evaluation of a spatial resolution metric for tomosynthesis: We-g-601-02," *Med. Phys.*, vol. 44, no. 6, p. 3261, 2017.
- [4] T. L. Vent, B. Barufaldi, and A. D. A. Maidment, "Simulation and experimental validation of high-resolution test objects for evaluating a next-generation digital breast tomosynthesis prototype," *Proc. SPIE*, vol. 2019, Mar. 2019, p. 21.
- [5] T. M. Morrison, P. Pathmanathan, M. Adwan, and E. Margerrison, "Advancing regulatory science with computational modeling for medical devices at the FDA's office of science and engineering laboratories," *Frontiers Med.*, vol. 5, p. 241, Sep. 2018.
- [6] A. Badano *et al.*, "In silico imaging clinical trials for regulatory evaluation: Initial considerations for VICTRE, a demonstration study," *Proc. SPIE*, vol. 10132, Mar. 2017, Art. no. 1013220.
- [7] P. Elangovan *et al.*, "Design and validation of realistic breast models for use in multiple alternative forced choice virtual clinical trials," *Phys. Med. Biol.*, vol. 62, no. 7, pp. 2778–2794, Apr. 2017.
- [8] E. Abadi *et al.*, "Virtual clinical trials in medical imaging: A review," *J. Med. Imag.*, vol. 7, no. 4, pp. 042805-1–042805-40, 2020.
- [9] A. Badano *et al.*, "Evaluation of digital breast tomosynthesis as replacement of full-field digital mammography using an in silico imaging trial," *JAMA Netw. Open*, vol. 1, no. 7, pp. 1–12, 2018.
- [10] D. D. Pokrajac, A. D. A. Maidment, and P. R. Bakic, "Optimized generation of high resolution breast anthropomorphic software phantoms," *Med. Phys.*, vol. 39, no. 4, p. 2290, 2012.
- [11] B. Barufaldi, P. R. Bakic, and A. D. A. Maidment, "Multiple-reader, multiple-case ROC analysis for determining the limit of calcification detection in tomosynthesis," *Proc. SPIE*, vol. 10948, p. 22, Mar. 2019.
- [12] R. J. Hooley, K. L. Greenberg, R. M. Stackhouse, J. L. Geisel, R. S. Butler, and L. E. Philpotts, "Screening US in patients with mammographically dense breasts: Initial experience with connecticut public act 09-41," *Radiology*, vol. 265, no. 1, pp. 59–69, Oct. 2012.
- [13] E. Sala, R. Warren, J. McCann, S. Duffy, N. Day, and R. Luben, "Mammographic parenchymal patterns and mode of detection: Implications for the breast screening programme," *J. Med. Screening*, vol. 5, no. 4, pp. 207–212, Dec. 1998.
- [14] K. Dembrower *et al.*, "Comparison of a deep learning risk score and standard mammographic density score for breast cancer risk prediction," *Radiology*, vol. 294, no. 2, pp. 265–272, Feb. 2020.
- [15] R. J. Acciavatti *et al.*, "Validation of the textural realism of a 3D anthropomorphic phantom for digital breast tomosynthesis," *Proc. SPIE*, vol. 10718, Jul. 2018, Art. no. 107180R.
- [16] R. A. Wahab, S.-J. Lee, B. Zhang, L. Sobel, and M. C. Mahoney, "A comparison of full-field digital mammograms versus 2D synthesized mammograms for detection of microcalcifications on screening," *Eur. J. Radiol.*, vol. 107, pp. 14–19, Oct. 2018.
- [17] S. P. Zuckerman *et al.*, "Implementation of synthesized two-dimensional mammography in a population-based digital breast tomosynthesis screening," *Radiol. J.*, vol. 281, no. 3, pp. 1–7, 2016.
- [18] P. Skaane *et al.*, "Two-view digital breast tomosynthesis screening with synthetically reconstructed projection images: Comparison with digital breast tomosynthesis with full-field digital mammographic images," *Radiol. J.*, vol. 271, no. 3, pp. 655–663, 2014.
- [19] E. A. Rafferty, J. M. Park, L. E. Philpotts, S. P. Poplack, and J. H. Sumkin, "Digital mammography and breast tomosynthesis compared with digital mammography alone: Results of a multicenter, multi-reader trial," *Radiology*, vol. 266, no. 1, pp. 104–113, 2013.

- [20] K. Bliznakova, S. Suryanarayanan, A. Karellas, and N. Pallikarakis, "Evaluation of an improved algorithm for producing realistic 3D breast software phantoms: Application for mammography," *Med. Phys.*, vol. 37, no. 11, pp. 5604–5617, Oct. 2010.
- [21] P. R. Bakic, D. D. Pokrajac, R. De Caro, and A. D. A. Maidment, *Realistic Simulation of Breast Tissue Microstructure in Software Anthropomorphic Phantoms* (Lecture Notes in Computer Science: Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics). Cham, Switzerland: Springer, 2014.
- [22] A.-A.-Z. Imran, P. R. Bakic, A. D. A. Maidment, and D. D. Pokrajac, "Optimization of the simulation parameters for improving realism in anthropomorphic breast phantoms," *Proc. SPIE*, vol. 10312, Mar. 2017, Art. no. 1013257.
- [23] A.-A.-Z. Imran, D. D. Pokrajac, A. D. A. Maidment, and P. R. Bakic, "Estimation of adipose compartment volume in CT images of a mastectomy specimen," *Proc. SPIE*, vol. 9783, Mar. 2016, Art. no. 97832O.
- [24] P. R. Bakic, D. D. Pokrajac, R. Caro, and A. D. A. Maidment, "Realistic simulation of breast tissue microstructure in software anthropomorphic phantoms," *Breast Imag.*, vol. 8539, pp. 348–355, Jun. 2014.
- [25] A. Avanaki, K. Espig, and T. Kimpe, "Location- and lesion-dependent estimation of mammographic background tissue complexity," *J. Med. Imag.*, vol. 4, no. 1, p. 15501, 2017.
- [26] L. R. Borges *et al.*, "Technical note: Noise models for virtual clinical trials of digital breast tomosynthesis," *Med. Phys.*, vol. 46, no. 6, pp. 2683–2689, Jun. 2019.
- [27] B. Barufaldi, T. L. Vent, R. J. Acciavatti, P. R. Bakic, P. B. Noel, and A. D. Maidment, "MRMC ROC analysis of calcification detection in tomosynthesis using computed super resolution and virtual clinical trials," *Proc. SPIE*, vol. 1151313, p. 58, May 2020.
- [28] A. Jonnalagadda *et al.*, "Evaluation of convolutional neural networks for search in $1/f^{2.8}$ filtered noise and digital breast tomosynthesis phantoms," *Proc. SPIE*, vol. 2020, Mar. 2020, Art. no. 1131617.
- [29] M. A. Lago, B. B. Barufaldi, P. R. Bakic, C. K. Abbey, A. D. A. Maidment, and M. P. Eckstein, "Foveated model observer to predict human search performance on virtual digital breast tomosynthesis phantoms," *Proc. SPIE*, vol. 11316, Mar. 2020, Art. no. 113160V.
- [30] C. K. Abbey, P. R. Bakic, D. D. Pokrajac, A. D. A. Maidment, M. P. Eckstein, and J. M. Boone, "Evaluation of non-Gaussian statistical properties in virtual breast phantoms," *J. Med. Imag.*, vol. 6, no. 2, p. 1, Jun. 2019.
- [31] P. R. Bakic, B. Barufaldi, D. Pokrajac, S. Weinstein, and A. Maidment, "Optimized simulation of breast anatomy for virtual clinical trials," in *Proc. 14th Int. Workshop Breast Imag. (IWBI)*, Jul. 2018, p. 73.
- [32] P. R. Bakic, D. Pokrajac, M. Feldman, and A. D. Maidment, "Simulation of sequential pathology images for the virtual clinical trials with rad-path correlation," in *Proc. 14th Int. Workshop Breast Imag. (IWBI)*, Jul. 2018, p. 74.
- [33] W. J. Van Der Laan, S. Green, and M. Sainz, "Screen space fluid rendering with curvature flow," in *Proc. 13D ACM SIGGRAPH Symp. Interactive 3D Graph. Games*, 2009, pp. 91–98.
- [34] F. Guo, J. Tang, and X. Xiao, "Foggy scene rendering based on transmission map estimation," *Int. J. Comput. Games Technol.*, vol. 2014, pp. 1–13, Jan. 2014.
- [35] R. Bridson, J. Houriham, and M. Nordenstam, "Curl-noise for procedural fluid flow," *ACM Trans. Graph.*, vol. 26, no. 3, p. 46, Jul. 2007.
- [36] W. Dong, X. Zhang, and C. Zhang, "Generation of cloud image based on Perlin noise," in *Proc. Int. Conf. Multimedia Commun.*, Aug. 2010, pp. 61–63.
- [37] R. Barnard and S. Ural, "Rendering translucency with Perlin noise," in *Proc. 3rd Int. Conf. Comput. Graph. Interact. Techn. Australasia South East Asia (GRAPHITE)*, 2005, pp. 131–134.
- [38] M. Dustler, P. Bakic, H. Petersson, P. Timberg, A. Tingberg, and S. Zackrisson, "Application of the fractal Perlin noise algorithm for the generation of simulated breast tissue," *Proc. SPIE*, vol. 2015, Mar. 2015, Art. no. 94123E.
- [39] K. Lång, M. Dustler, and H. Petersson, "Binary implementation of fractal Perlin noise to simulate fibroglandular breast tissue," *Proc. SPIE*, vol. 1057357, Mar. 2018, p. 193.
- [40] S. Worley, "A cellular texture basis function," in *Proc. 23rd Annu. Conf. Comput. Graph. Interact. Techn. (SIGGRAPH)*, 1996, pp. 291–294.
- [41] M. Abdolhoseini, M. G. Kluge, F. R. Walker, and S. J. Johnson, "Neuron image synthesizer via Gaussian mixture model and perlin noise," in *Proc. IEEE 16th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2019, pp. 530–533.
- [42] H.-J. Bae *et al.*, "A Perlin noise-based augmentation strategy for deep learning with small data samples of HRCT images," *Sci. Rep.*, vol. 8, no. 1, pp. 1–7, Dec. 2018.
- [43] W. Burger and M. J. Burge, *Principles of Digital Image Processing: Advanced Methods (Supplemental Material)*, 3rd ed. London, U.K.: Springer, 2013.
- [44] K. Perlin, "Image synthesizer," *Comput. Graph.*, vol. 19, no. 3, pp. 287–296, 1985.
- [45] K. Perlin, "Improving noise," *ACM Trans. Graph.*, vol. 21, no. 3, pp. 681–682, 2002.
- [46] K. G. Metheany, C. K. Abbey, N. Packard, and J. M. Boone, "Characterizing anatomical variability in breast CT images," *Med. Phys.*, vol. 35, no. 10, pp. 4685–4694, Sep. 2008.
- [47] L. Chen, J. M. Boone, A. Nosratieh, and C. K. Abbey, "NPS comparison of anatomical noise characteristics in mammography, tomosynthesis, and breast CT images using power law metrics," *Proc. SPIE*, vol. 7961, Mar. 2011, Art. no. 79610F.
- [48] L. Cockmartin, H. Bosmans, and N. W. Marshall, "Comparative power law analysis of structured breast phantom and patient images in digital mammography and breast tomosynthesis," *Med. Phys.*, vol. 40, no. 8, Jul. 2013, Art. no. 081920.
- [49] A. E. Burgess, F. L. Jacobson, and P. F. Judy, "Human observer detection experiments with mammograms and power-law noise," *Med. Phys.*, vol. 28, no. 4, pp. 419–437, Apr. 2001.
- [50] C. K. Abbey, A. Nosratieh, J. Sohl-Dickstein, K. Yang, and J. M. Boone, "Non-Gaussian statistical properties of breast images," *Med. Phys.*, vol. 39, no. 11, pp. 7121–7130, Nov. 2012.
- [51] B. Barufaldi, P. R. Bakic, D. Higginbotham, and A. D. A. Maidment, "OpenVCT: A GPU-accelerated virtual clinical trial pipeline for mammography and digital breast tomosynthesis," *Proc. SPIE*, vol. 1057, Mar. 2018, Art. no. 1057358.
- [52] K. Perlin and E. M. Hoffert, "Hypertexture," in *Proc. 16th Annu. Conf. Comput. Graph. Interact. Techn., SIGGRAPH*, 1989, pp. 253–262.
- [53] A. Mertins, "Wavelet transform," in *Signal Analysis*. Chichester, NY, USA: Wiley, 2003, pp. 210–264.
- [54] A. F. Laine and X. Zong, "A multiscale sub-octave wavelet transform for de-noising and enhancement," in *Wavelet Applications in Signal and Image Processing IV*. Denver, CO, USA: SPIE, 1996, pp. 238–249.
- [55] W. Burger, "Imagingbook-common.jar: A compiled collection of Java files," Tech. Rep., 2013.
- [56] B. Barufaldi, P. Bakic, and A. D. A. Maidment. (2020). Open Virtual Clinical Trials. *OpenVCT: A GPU-Accelerated Virtual Clinical Trial Pipeline for Mammography and Digital Breast Tomosynthesis*. Accessed: Mar. 10, 2019. [Online]. Available: <https://sourceforge.net/projects/openvct/>
- [57] M. A. Lago, A. D. A. Maidment, and P. R. Bakic, "Modelling of mammographic compression of anthropomorphic software breast phantom using FEBio," in *Proc. Int. Symp. Comput. Methods Biomech. Biomed. Eng.* Salt Lake, UT, USA: European Society of Biomechanics, 2013, pp. 487–488.
- [58] T. Pietzsch, S. Preibisch, P. Tomank, and S. Saalfeld, "IMG lib 2—Generic image processing in Java," *Bioinformatics*, vol. 28, no. 22, pp. 3009–3011, 2012.
- [59] J. R. Rumble, D. M. Bickham, and C. J. Powell, "The NIST X-ray photoelectron spectroscopy database," *Surf. Interface Anal.*, 1992.
- [60] R. L. Siddon, "Fast calculation of the exact radiological path for a three-dimensional CT array," *Med. Phys.*, vol. 12, no. 2, pp. 252–255, Mar. 1985.
- [61] S. S. J. Feng and I. Sechopoulos, "Clinical digital breast tomosynthesis system: Dosimetric characterization," *Radiology*, vol. 263, no. 1, pp. 35–42, Apr. 2012.
- [62] *ICRU Report 44—Tissue Substitutes in Radiation Dosimetry and Measurement*, ICRU, Bethesda, MD, USA, 1988.
- [63] J. Kuo, P. A. Ringer, S. G. Fallows, P. R. Bakic, A. D. A. Maidment, and S. Ng, "Dynamic reconstruction and rendering of 3D tomosynthesis images," *Proc. SPIE*, vol. 796116, Mar. 2011, pp. 796116-1–796116-11.
- [64] J. H. Chui, D. D. Pokrajac, A. D. A. Maidment, and P. R. Bakic, "Roadmap for efficient parallelization of breast anatomy simulation," *Proc. SPIE*, vol. 8313, Mar. 2012, p. 83134T.
- [65] J. G. Daugman, "Two-dimensional spectral analysis of cortical receptive field profiles," *Vis. Res.*, vol. 20, no. 10, pp. 847–856, Jan. 1980.
- [66] J. P. Jones and L. A. Palmer, "An evaluation of the two-dimensional Gabor filter model of simple receptive fields in cat striate cortex," *J. Neurophysiol.*, vol. 58, no. 6, pp. 1233–1258, Dec. 1987.